

MICROELECTROMECHANICAL SYSTEMS AND SENSORS

MEMS, sensors, and the measurement-system architecture

Lecture 1 of 16 · 80 minutes · Module A: Foundations

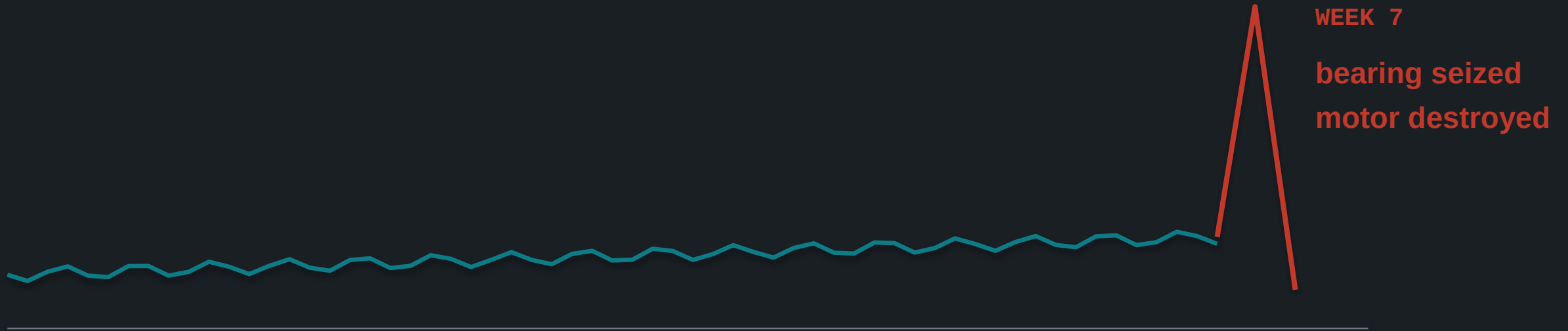
Bachelor programme · Electrical Engineering · 7th semester

Institute of Energy and Electrical Engineering

The motor that died under observation

A 1500 rpm induction motor. A vibration monitor. Six weeks of data.

Six weeks of calm



week 1

20 Hz · amplitude slowly rising · read as normal load variation

Nothing failed

The post-mortem checked every component

The accelerometer

✓ met every number in its datasheet

The circuit board

✓ no fault found

The firmware

✓ no bug found

The data logger

✓ logged every single sample

The engineer

✓ looked at the screen every week

Where was the fault?

- A The accelerometer was too cheap for industrial use
- B There was a firmware bug nobody found
- C Nobody specified the sample rate against the frequency of the fault
- D Bearing degradation cannot be detected by vibration
- E The maintenance team ignored a rising trend

Commit now. No discussion. I will show you the answer in the last ten minutes of this lecture — and I want to see whether you change your mind.

What this course is

and, just as importantly, what it is not

THIS COURSE IS

Choosing, interfacing, calibrating and validating real sensors inside real embedded systems.

IT IS NOT

A cleanroom fabrication course.
We visit fabrication once, in Lecture 4, only as deep as packaging and drift require.

The design principle of all sixteen lectures: **understand the measurand → select the sensor → interface it → acquire valid data → calibrate it → process or fuse it → validate it in a system.**

By the end of today you can...

- 01 **DRAW** the path from a physical quantity to a logged number, and say where information dies
- 02 **CLASSIFY** an unfamiliar device: sensor, transducer or actuator — and name its measurand
- 03 **PREDICT** from scaling laws why MEMS devices are fast — and what gets worse when they shrink
- 04 **CONVERT** a vague request into a measurement specification you could order against

Notice that all four are verbs. None of them is “know about”.

The chain

A measurement is not a component. It is a path — and a path is only as truthful as its worst link. We are going to build that path one box at a time.

Three words we will use precisely

because from here on the difference matters

SENSOR

Converts a physical quantity into a signal you can read.

MEMS microphone
sound pressure → voltage

TRANSDUCER

Converts energy from one form to another. A sensor is one kind.

loudspeaker, strain gauge,
piezo disc – either
direction

ACTUATOR

Converts a signal into a physical action on the world.

MEMS mirror
voltage → beam deflection

Every actuator in this course is also a measurement problem: you only know it acted if you sense the result.

Thirty seconds

Sensor, transducer or actuator? And what is the measurand?

**MEMS
microphone**

**Piezo
buzzer**

**Strain
gauge**

**MEMS
mirror**

?

?

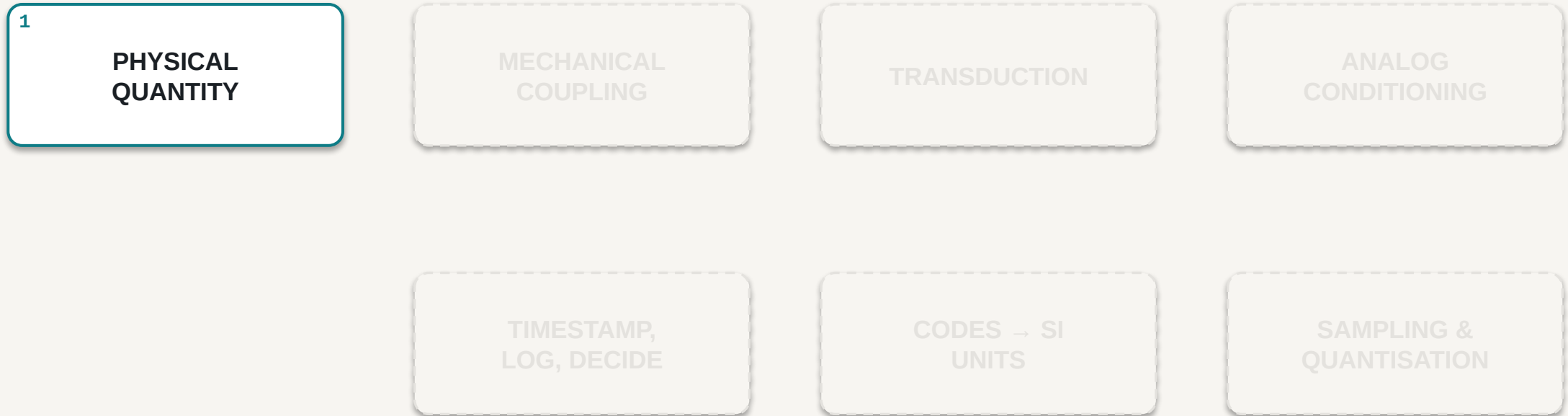
?

?

Turn to your neighbour. Four devices, thirty seconds. Say the measurand out loud.

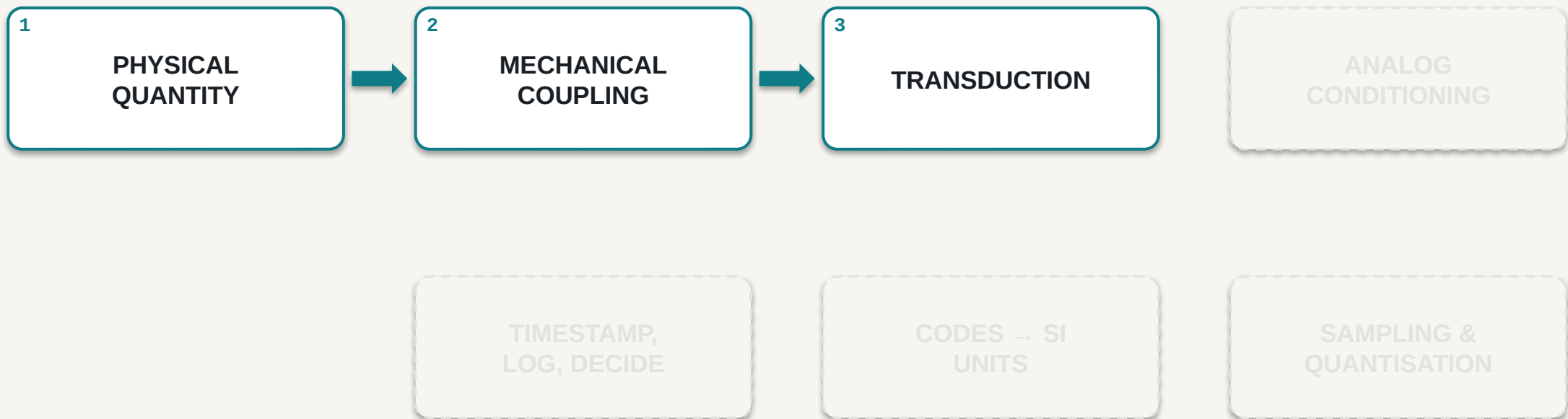
One of these four is passive — it changes a resistance and needs an excitation current before it says anything at all. Which one, and why does that matter?

The measurand



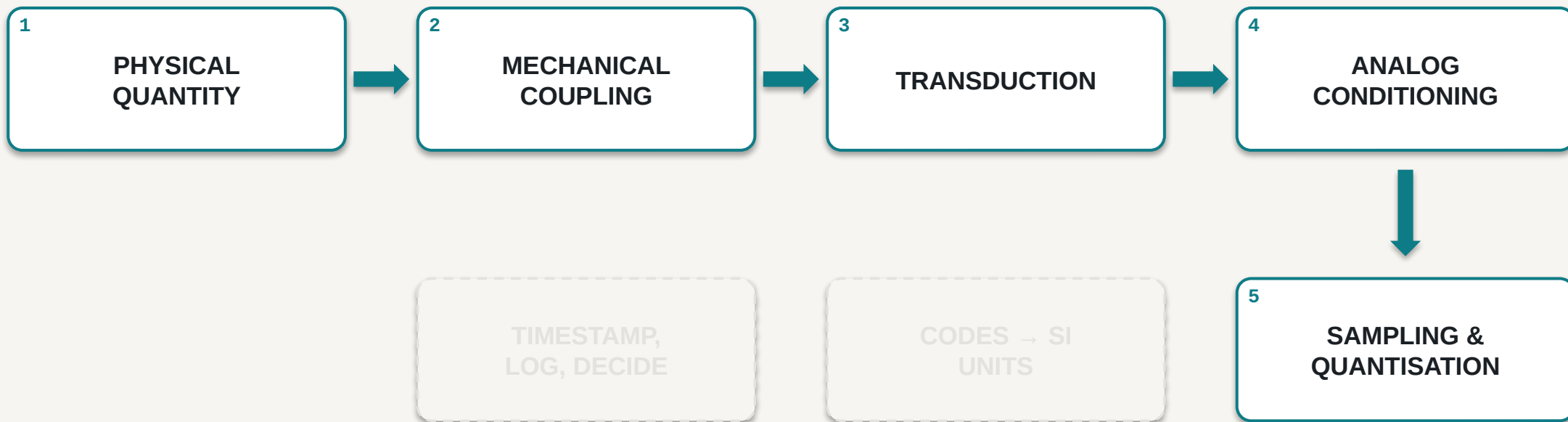
It does not start at the sensor

The measurement chain



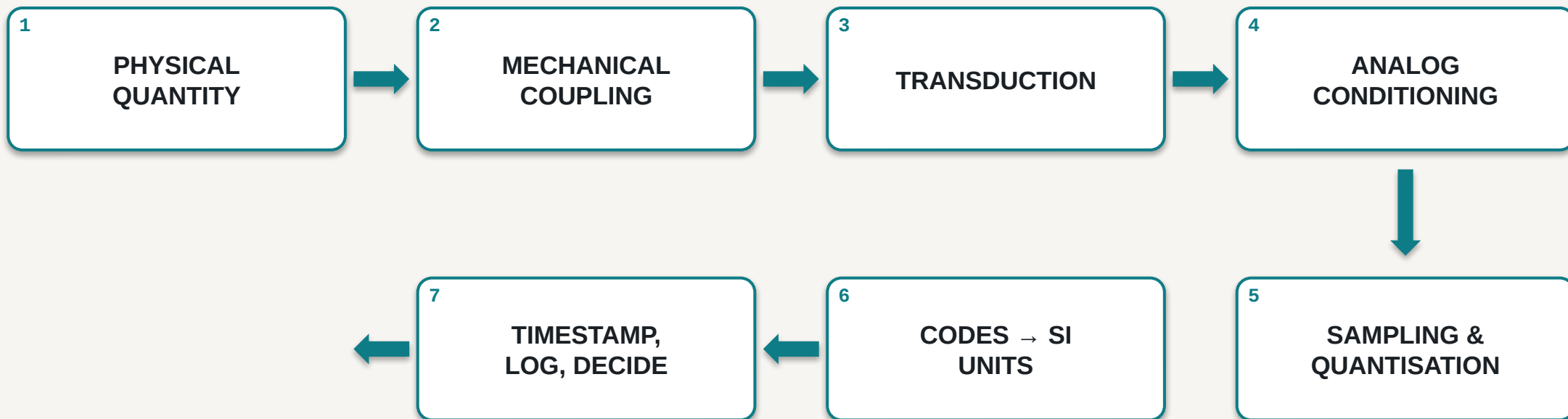
Through the electronics

The measurement chain



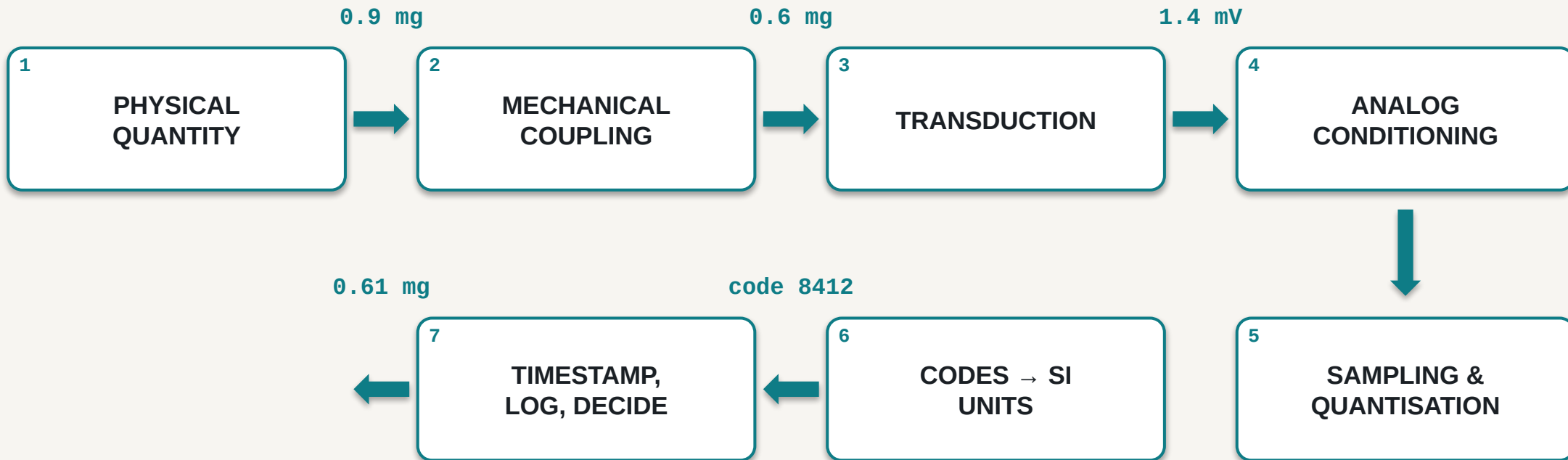
The full chain

The measurement chain



The same chain, with real numbers

One 0.9 mg vibration, all the way to a decision



Notice stage 2: the mechanical path already lost a third of the signal — before a single electron moved.

Where the truth dies

and whether you can ever get it back

Stage	What is lost	Recoverable later?
2 Mechanical coupling	amplitude and phase, frequency-dependent	No — measure it yourself
3 Transduction	cross-axis leakage, nonlinearity, temperature drift	Partly — by calibration
4 Conditioning	saturation clips peaks; wrong filter kills the signal	No
5 Sampling	everything above half the sample rate, aliased in	Never
5 Quantisation	detail below one LSB	Partly — averaging
6 Scaling to units	accuracy, if the sensitivity constant is wrong	Yes — recompute
7 Timestamping	the time axis, and therefore all frequency content	No

Three of these are permanent. Design decisions, not calibration problems.

An accelerometer with 0.061 mg resolution is bolted to a motor housing through a 5 mm rubber pad. Which stage has already destroyed the accuracy of the reported number?

- A** Digitisation — 16 bits cannot support three decimal places
- B** Unit conversion — mg was never converted to m/s²
- C** Mechanical coupling — the rubber pad filters the vibration before any electronics see it
- D** No stage — 0.061 mg resolution guarantees the digits are meaningful

Vote alone first. Then find someone who disagrees with you.

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The measurement chain begins before the sensor. If the mechanical path lies to the proof mass, no downstream precision recovers the truth.

The chain begins before the sensor.

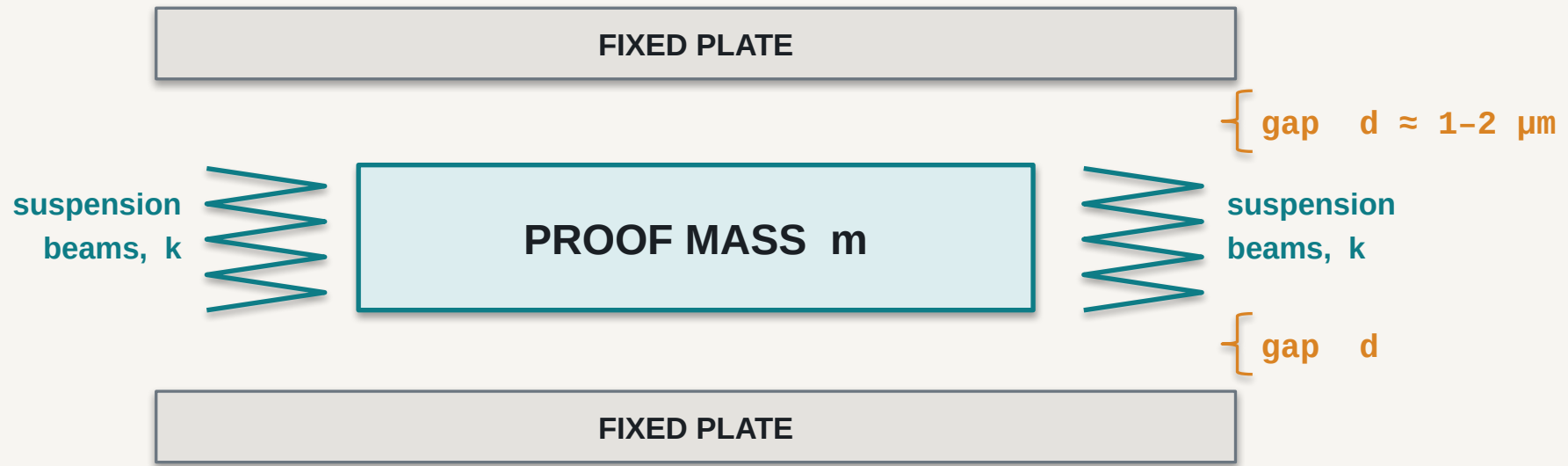
Your instrument includes the bolt, the bracket and the housing. None of them are in the datasheet, and all of them are yours.

Why micro is different

You have spent three years building intuition on machines you can hold.
At the micrometre scale, half of that intuition inverts. Here is which half.

What is actually inside

A capacitive MEMS accelerometer, in cross-section



Acceleration moves the mass. Movement changes both gaps — one grows, one shrinks — and the differential capacitance $C = \epsilon A/d$ is the signal.

That is the whole device — everything in the datasheet is a consequence of m , k and d .

Shrink every dimension by the same factor

Isotropic scaling: length, width, thickness — all by factor s

$$m \propto L^3$$

mass scales with volume

$$k \propto L^1$$

beam stiffness $k = E \cdot w \cdot t^3 / 4L^3$

They do not scale together. That single fact is the whole lecture.

Substitute w , t and L all by the same factor s into $k = E \cdot w \cdot t^3 / 4L^3$ and you get $s \cdot s^3 / s^3 = s$. Stiffness falls linearly. Mass falls cubically.

So the resonant frequency rises

$$f_0 = (1/2\pi) \cdot \sqrt{(k/m)} \propto \sqrt{(L^1 / L^3)} = \sqrt{(L^{-2})} = 1 / L$$

Shrink the device by 10 → its resonant frequency rises by 10

Device	Typical size	Resonance
Bridge span	100 m	≈ 0.2 Hz
Tuning fork	10 cm	≈ 440 Hz
MEMS accelerometer	300 μm	≈ 5 kHz
MEMS gyroscope drive	100 μm	≈ 20 kHz
MEMS RF resonator	10 μm	≈ 100 MHz+

This is why MEMS can measure fast things.

A sensor cannot report signals near its own resonance — so a high f_0 is what buys you bandwidth.

A MEMS accelerometer's proof mass and suspension beams are all scaled down by a factor of 10. Which statement is true?

- A** Resonant frequency rises $\times 10$, and thermomechanical noise gets worse
- B** Resonant frequency falls $\times 10$, and noise improves because it is smaller
- C** Resonant frequency is unchanged — mass and stiffness both shrink, so the ratio is constant
- D** Resonant frequency rises $\times 10$, and noise improves because there is less material to vibrate

The hardest question of the lecture. Use the two exponents on the board, not your intuition.

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- B Resonant frequency falls $\times 10$, and noise improves because it is smaller
- C Resonant frequency is unchanged — mass and stiffness both shrink, so the ratio is constant
- D Resonant frequency rises $\times 10$, and noise improves because there is less material to vibrate

Thermomechanical noise $\propto \sqrt{(4kBT \cdot \omega_0 / mQ)}$ — it rises as the proof mass falls. You buy bandwidth and you pay in noise floor.

Smaller is not uniformly better

The honest ledger of shrinking a device

GETS BETTER

- ▶ Bandwidth — $f_0 \propto 1/L$
- ▶ Response time and thermal settling
- ▶ Power per device
- ▶ Cost per device at volume
- ▶ Batch fabrication: thousands per wafer

GETS WORSE

- ▶ Thermomechanical noise floor $\propto 1/\sqrt{m}$
- ▶ Stiction — surface forces beat body forces
- ▶ Sensitivity to packaging stress
- ▶ Temperature drift and offset stability
- ▶ Contamination: a dust speck is fatal

Surface area / volume $\propto 1 / L$ — at small scale the world is all surface.

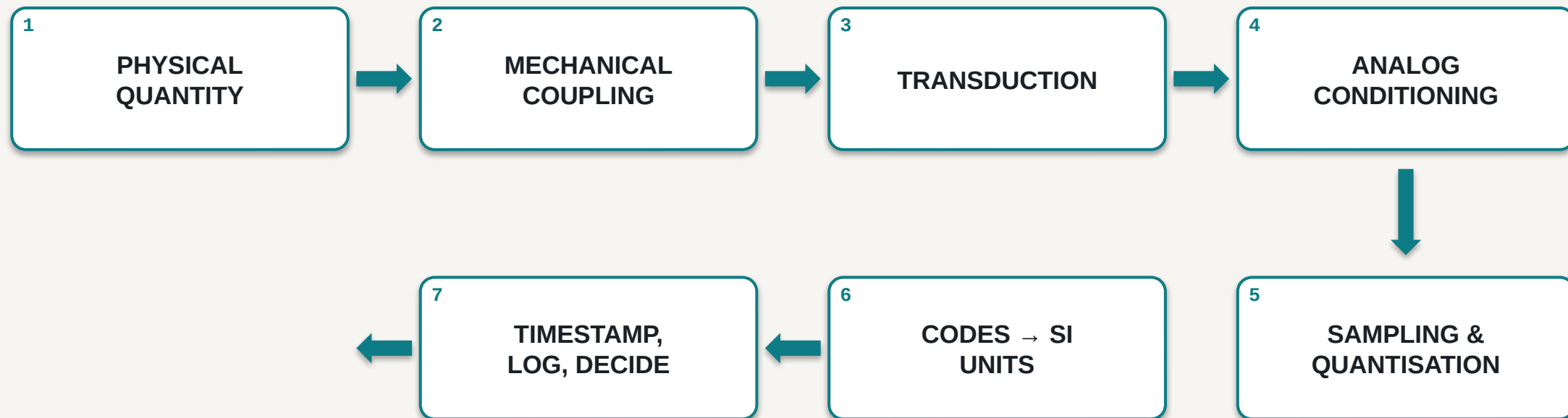
Slides off. Paper out.

- 1 Stand up. Actually stand.
- 2 In pairs, one sheet: redraw the measurement chain from memory. Every box, every arrow. 90 seconds.
- 3 Mark an X on every box where the motor's number could have been ruined.

Keep this sheet. You will use it at the bench in Week 2.

Check your sheet

Correct in a different colour — do not rewrite it



Two boxes are forgotten more than any others: mechanical coupling, and timestamping.

Both are invisible in a datasheet. Both are yours.

From a wish to a specification

Nobody will ever hand you a specification. They will hand you a sentence like the one on the next slide, and expect an instrument at the end of it.

**“Just tell me if the motor
is vibrating too much.”**

This is what a request actually looks like. It contains no quantity, no range, no bandwidth, no accuracy and no environment. It is not yet an engineering problem.

Seven questions that turn a wish into a specification

QUANTITY

What physical variable, in what units?

RANGE

Smallest and largest value that must be reported?

RESOLUTION

Smallest change that must be distinguishable?

ACCURACY

How wrong is a reading allowed to be — and over what temperature?

BANDWIDTH

How fast does it change? What is the highest frequency that matters?

ENVIRONMENT

Temperature, humidity, vibration, EMI, supply, power budget?

OUTPUT

Who consumes the number, in what form, how often, timestamped how?

One of these seven killed the motor. You already know which one.

The same request, specified

Now it is something you can order against

	Specification	Where the number came from
Quantity	housing acceleration, m/s ² RMS	vibration is what a bearing radiates
Range	±16 g	impact transients, not steady vibration
Resolution	≤ 5 mg	earliest detectable fault ≈ 30 mg
Accuracy	±5 % of reading, 0–70 °C	trend detection, not absolute metrology
Bandwidth	≥ 4 kHz, anti-aliased	bearing signature spans 1–4 kHz
Environment	0–70 °C, 24 V rail, oily, EMI from a VFD	the motor's actual cabinet
Output	RMS + spectrum, 1 Hz, timestamped ±1 ms	so a frequency can be computed

Seven lines. This is the deliverable — not a part number.

Your turn

Four minutes, in pairs — write the seven lines

“The drone keeps drifting up and down. Make it hold its altitude properly.”

Start here

What is the physical quantity?
It is probably not altitude.

The hard one

What bandwidth? How fast does
a drone's altitude actually
change?

The trap

What is the RANGE — and is it
the
same as the RESOLUTION you
need?

You will not finish all seven lines. Get three right.

Bearing fault energy appears between 1 and 4 kHz. Which single specification line decides whether your system can work at all?

- A** Full-scale range $\geq \pm 16$ g
- B** Bandwidth ≥ 4 kHz, sample rate ≥ 8 kHz, with anti-alias filtering
- C** Resolution ≤ 0.1 mg
- D** Operating temperature up to 85 °C

Every one of these four is a real requirement. Only one of them, if you get it wrong, makes the system incapable rather than merely imperfect.

Bearing fault energy appears between 1 and 4 kHz. Which single specification line decides whether your system can work at all?

A Full-scale range $\geq \pm 16$ g

B Bandwidth ≥ 4 kHz, sample rate ≥ 8 kHz, with anti-alias filtering

C Resolution ≤ 0.1 mg

D Operating temperature up to 85 °C

A, C and D are all genuine requirements — get them wrong and the system is imperfect. Get B wrong and the system is blind.

The semester project

Deliverable 1 is due before Lecture 3

WHAT YOU WILL BUILD

A multi-sensor embedded system that measures something real, calibrated, filtered, timestamped, validated against stated requirements — and honest about its own limitations.

DELIVERABLE 1 · ONE PAGE

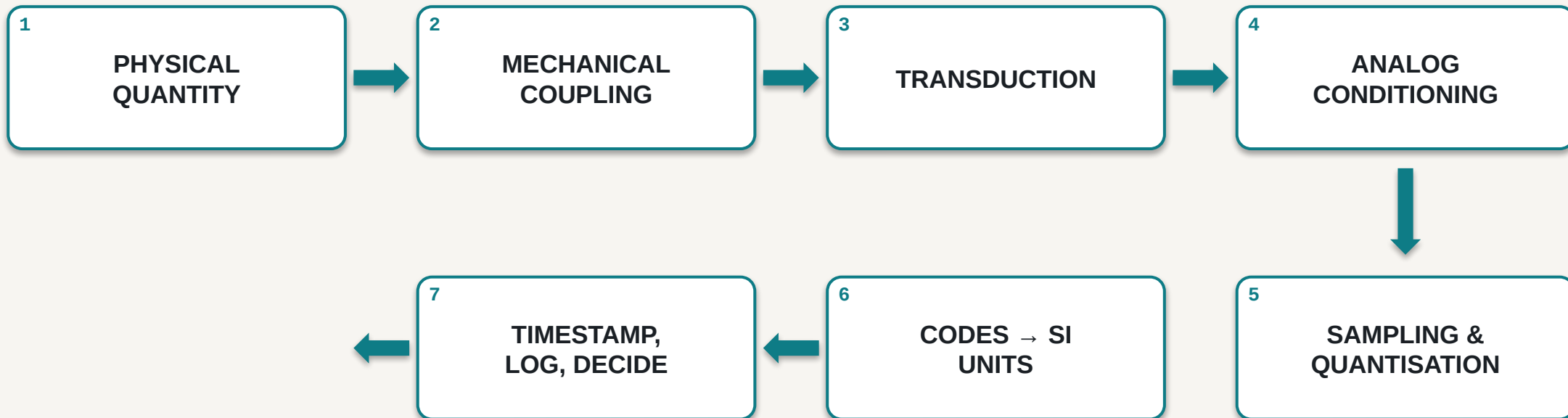
The seven specification lines, for a measurement problem you choose.

No part numbers. No circuit. No code. Just the seven lines — each with the reason it has that value.

Teams of 2–3. Same teams as the laboratory. Choose your measurand this week.

Back to the motor

Before I tell you — commit, physically



Put your finger on the box where you now think this failed. Everyone. I am looking.

The three numbers from the post-mortem

1520 Hz

the bearing fault signature

100 Hz

the sample rate somebody chose

none

anti-alias filter fitted

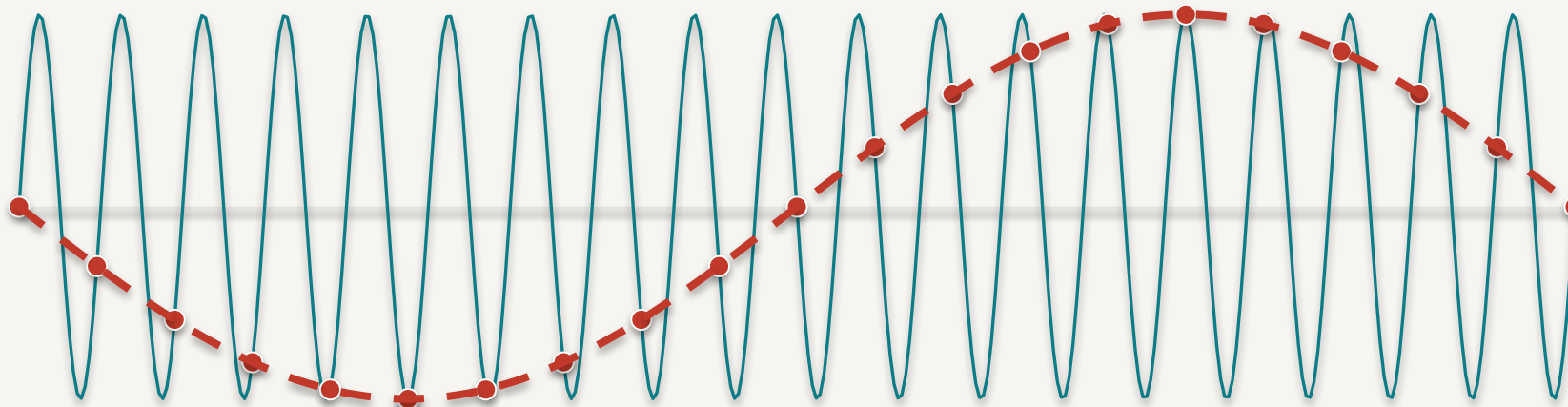
Sampling at 100 Hz, everything above 50 Hz is folded down into the band — and with no filter in front of the ADC, there was nothing to stop it.

Nyquist: to see 1520 Hz you need to sample above 3040 Hz. They sampled 30 times too slowly.

Where the 20 Hz line came from

Drawn at 19 cycles per 20 samples so you can see it — the motor's ratio was 1520:100, identical arithmetic

the real vibration · 19 cycles



what the samples reconstruct · 1 cycle

red dots =
the only instants
the ADC looked

$$\begin{array}{l} | \quad 1520 - 15 \times 100 \quad | \\ = \quad 20 \text{ Hz} \end{array}$$

The monitor was faithfully displaying the destruction of the bearing - relabelled as a slow load variation.

**Nothing was broken.
Every part met spec.
The system was never specified.**

Aliasing is permanent. No filter afterwards, no clever software, no machine learning recovers a frequency you failed to sample.

What was missing

One line, in one document, that nobody wrote

**Bandwidth: ≥ 4 kHz, anti-aliased. Sample rate:
 ≥ 8 kHz.**

One line. One motor.

This is also your answer to Poll 4 — option **B**. And it is the specification line you were asked to write fifteen minutes ago.

Lecture 3 is about how that line becomes an ADC configuration. Laboratory 2 is where you make aliasing happen with your own hands.

Four things to keep

01 **A measurement is a chain, and it begins before the sensor.**

Bolt, bracket and housing are part of your instrument.

02 **Some losses are permanent.**

Aliasing and a missing timestamp cannot be repaired downstream.

03 **Shrinking a device is a trade, not an improvement.**

$k \propto L$, $m \propto L^3$, $f_0 \propto 1/L$ — bandwidth up, noise floor up too.

04 **The specification is the deliverable.**

Seven lines, each with a reason. Part numbers come afterwards.

Exit ticket

Anonymous. Hand it in at the door — both halves.

1 · ONE SPECIFICATION LINE

A drone must hold altitude to ± 0.5 m.

Write ONE specification line for its pressure sensor — the line you think matters most, with its number.

This is the first time this course has been taught. Your muddiest points genuinely change what happens next week.

2 · MUDDIEST POINT

What is the one thing from today you are least sure about?

One sentence. I will answer the three most common at the start of Lecture 2.

Week 2: Laboratory 1

Datasheet to data — and yes, it is before the theory

At the bench you will	Which chain stage
Read supply and logic levels from the datasheet BEFORE applying power	before stage 1
Wire one I ² C sensor to the Nucleo board, with pull-ups	stage 3 → 5
Prove the device is the device you think it is (device-ID register)	stage 5
Configure a full-scale range and an output data rate	stage 5
Convert raw two's-complement codes to SI units — by hand	stage 6
Prove the number is true: one axis must read ≈ 9.81 m/s ² at rest	stage 6

No programming. The boards are already flashed — you drive the sensor from a serial console.

**The pre-lab sheet is a gate: no signed pre-lab, no power to your board.
It carries a one-page I²C primer and the console reference — read it before Tuesday.**

Lecture 2

Sensor specifications and datasheet-based selection

Bring a laptop or a printed datasheet. We are going to find a number that decides a design — and it will not be on the front page.

BEFORE NEXT WEEK

- 1 · Formative quiz (6 questions, ungraded) — posted tonight
- 2 · Pre-lab sheet for Laboratory 1, including the I²C primer
- 3 · Project deliverable 1: your seven specification lines
- 4 · Reading: Fraden, Handbook of Modern Sensors — Ch. 1–2